

The $C(K)$ -Predual Subcase of Question 6.7 in arXiv:2604.18157

fa_banach_001, agent_lane_15

2026-06-29

Source and Question

The source paper is Ondrej–Spurny, *Operators on injective tensor products of L_1 -preduals*, arXiv:2604.18157v1 [1]. Let X be an L_1 -predual and E, F Banach spaces. If

$$T : X \widehat{\otimes}_\varepsilon E \rightarrow F$$

is strongly bounded, the paper constructs a canonical extension

$$S : C(B_{X^*}, E) \rightarrow F, \quad Sf = \int_{B_{X^*}} f dG,$$

where G is the representing measure of T .

Theorem 6.5(b) proves that, for pseudo weakly compact and Right completely continuous operators, S having the property implies T has it; the converse is proved when X is separable. The paper asks whether the separability assumption can be removed.

such that the values of G are compact, yet T is not unconditionally converging.

Question 6.7. *We do not know whether (b) in Theorem 6.5 holds without the assumption of separability of X .*

Result

Theorem 1. *Let K be a compact Hausdorff space and let $X = C(K)$, over the real or complex scalar field $\mathbb{F}$. Let E, F be Banach spaces and let*

$$T : C(K) \widehat{\otimes}_\varepsilon E \simeq C(K, E) \rightarrow F$$

be a strongly bounded operator. Let

$$S : C(B_{X^*}, E) = C(B_{M(K)}, E) \rightarrow F$$

be the canonical extension from Ondrej–Spurny.

Then, for each of the following two operator classes,

pseudo weakly compact, Right completely continuous,

T belongs to the class if and only if S belongs to the class.

Thus Question 6.7 has an affirmative answer for all $C(K)$ -preduals, with no separability or metrizability assumption on K .

Proof Intuition

For a general L_1 -predual, the source paper's boundary projection D need only send continuous functions to Borel homogeneous affine functions. Separability is used to replace this Borel object by continuous affine selections. For $X = C(K)$, however, D has a concrete continuous formula: first average a function over the homogeneous extreme boundary $\{\alpha\delta_t : \alpha \in S_{\mathbb{F}}, t \in K\}$, then integrate the result against $\mu \in B_{M(K)}$. This gives a norm-one projection from $C(B_{M(K)}, E)$ onto the copy of $C(K, E)$. The canonical extension therefore factors as $S = TP$. Since bounded maps preserve Right-null and Right-Cauchy sequences, the desired preservation properties are immediate.

The Projection

Let m denote normalized Haar measure on $S_{\mathbb{F}}$ (counting measure on $\{-1, 1\}$ in the real case). For $f \in C(B_{M(K)}, E)$, define

$$(Rf)(t) = \int_{S_{\mathbb{F}}} \alpha^{-1} f(\alpha\delta_t) dm(\alpha), \quad t \in K,$$

and then

$$(Pf)(\mu) = \int_K (Rf)(t) d\mu(t), \quad \mu \in B_{M(K)}.$$

Here the second integral is the usual vector integral of a continuous E -valued function against the scalar regular measure μ .

Lemma 1. *P is a norm-one projection from $C(B_{M(K)}, E)$ onto $A_{\text{hom}}(B_{M(K)}, E)$, which is the canonical copy of $C(K) \widehat{\otimes}_{\varepsilon} E \simeq C(K, E)$.*

Proof. The map $R : C(B_{M(K)}, E) \rightarrow C(K, E)$ is bounded with $\|R\| \leq 1$, since it is an average of restrictions to the compact set $\{\alpha\delta_t : \alpha \in S_{\mathbb{F}}, t \in K\}$. If $g \in C(K, E)$, its affine homogeneous extension to $B_{M(K)}$ is

$$\widehat{g}(\mu) = \int_K g d\mu.$$

Thus $Pf = \widehat{Rf}$ belongs to $A_{\text{hom}}(B_{M(K)}, E)$ and $\|P\| \leq 1$. If $f = \widehat{g}$ is already affine homogeneous, then

$$f(\alpha\delta_t) = \alpha g(t),$$

so $Rf = g$ and $Pf = \widehat{g} = f$. Hence P is a projection onto $A_{\text{hom}}(B_{M(K)}, E)$. Its norm is one unless the space is trivial, because it is the identity on constants of the form $\widehat{g}$ with $\|g\| = 1$. $\square$

Lemma 2. *For the canonical extension S of T , one has $S = TP$.*

Proof. It is enough to check after applying an arbitrary $y^* \in F^*$. Let G_{y^*} be the scalarized E^* -valued representing measure of T . In the source paper, scalar components of G_{y^*} belong to the antihomogeneous boundary measure space associated with X .

For $X = C(K)$, the source paper's projection D on scalar measures has the following explicit function-side adjoint: for scalar $\varphi \in C(B_{M(K)})$,

$$(D\varphi)(\mu) = \int_K \left(\int_{S_{\mathbb{F}}} \alpha^{-1} \varphi(\alpha\delta_t) dm(\alpha) \right) d\mu(t).$$

This is exactly the scalar version of P . Since G_{y^*} is already antihomogeneous, the source identity $D\nu = \nu$ for antihomogeneous boundary measures gives

$$\int_{B_{M(K)}} \varphi d\nu = \int_{B_{M(K)}} D\varphi d\nu.$$

Applying this to all scalar functions $e^* \circ f$, $e^* \in E^*$, yields

$$y^*(Sf) = \int_{B_{M(K)}} f dG_{y^*} = \int_{B_{M(K)}} Pf dG_{y^*} = y^*(T(Pf)).$$

Since y^* was arbitrary, $Sf = T(Pf)$. □

Proof of Theorem 1

Proof. The implication $S \Rightarrow T$ is the easy half already present in Theorem 6.5(b) of [1], since T is the restriction of S to $A_{\text{hom}}(B_{M(K)}, E)$.

Conversely assume first that T is pseudo weakly compact. If (f_n) is Right-null in $C(B_{M(K)}, E)$, then (Pf_n) is Right-null in $A_{\text{hom}}(B_{M(K)}, E)$, because P is bounded and bounded operators pull weakly compact subsets of the target dual back to weakly compact subsets of the domain dual. Hence

$$\|Sf_n\| = \|T(Pf_n)\| \rightarrow 0.$$

Thus S is pseudo weakly compact.

The Right completely continuous case is identical. If (f_n) is Right-Cauchy in $C(B_{M(K)}, E)$, then (Pf_n) is Right-Cauchy in $A_{\text{hom}}(B_{M(K)}, E)$. Since T is Right completely continuous, $(T(Pf_n))$ converges in the Right topology of F . By Lemma 2, this is exactly (Sf_n) . Therefore S is Right completely continuous. □

Verification and Limitations

The proof was checked against the source TeX around Lemma 2.7, Theorem 6.5, and Question 6.7. Local run indexes were searched for arXiv:2604.18157, the paper title, Question 6.7, pseudo weakly compact, Right completely continuous, $C(K)$, and injective tensor product terms. The run already contains one partial packet for metrizable boundary support and one no-promotion full-push attempt; no packet for this $C(K)$ -predual factorization subcase was found.

This is a genuine subcase only. It does not settle arbitrary L_1 -preduals, because for a general L_1 -predual the projection D need not be continuous on $C(B_{X^*}, E)$. That continuity is the special feature of the concrete $C(K)$ -predual representation.

Human Review Recommendation

The main point to verify is Lemma 2: in the case $X = C(K)$, the concrete projection P above is the function-side form of the antihomogeneous boundary projection D used by Ondrej–Spurný. Once that identification is accepted, the Right-topology argument is formal.

References

- [1] S. Ondrej and J. Spurny, *Operators on injective tensor products of L_1 -preduals*, arXiv:2604.18157v1, 2026.